WORKSHEET 1.11

Unit 1 Test: Limits and Continuity

Unit 1: Limits and Continuity · 20 questions · 45 minutes

Instructions

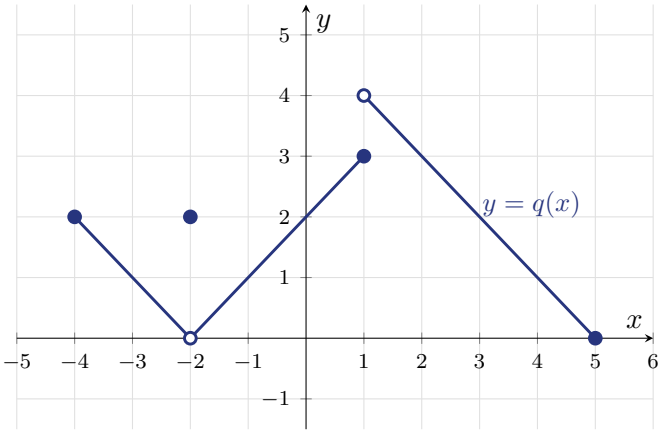
- **Part A** (questions 1–15): 32 minutes, **no calculator**.
- **Part B** (questions 16–20): 13 minutes, a graphing calculator is required.
- Each question has exactly one correct answer. Circle one letter. There is no penalty for a wrong answer, so answer every question.
- Questions 2, 3 and 14 refer to the graph of q printed with question 2.

Part A · No calculator

1. $\lim\limits_{x\to 2} \dfrac{x^2-4}{x^2+x-6} =$

   (A) $\dfrac{4}{5}$ (B) 0 (C) 1 (D) does not exist

2. The graph of the function q, defined on $[-4, 5]$, is shown.



   What is $\lim\limits_{x\to -2} q(x)$?

   (A) 2 (B) 0 (C) −2 (D) does not exist

3. For the function q graphed above, $\lim\limits_{x\to 1} q(x)$ is

   (A) 3 (B) 4 (C) $\dfrac{7}{2}$ (D) nonexistent

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4. $\lim_{x\to 0} \dfrac{\sin(3x)}{4x} =$

(A) 0 (B) $\frac{3}{4}$ (C) $\frac{4}{3}$ (D) 1

5. $\lim_{x\to \infty} \dfrac{6x^2-1}{2x^2+5x} =$

(A) 0 (B) 3 (C) 6 (D) ∞

6. $\lim_{x\to -\infty} \dfrac{\sqrt{16x^2+3}}{2x} =$

(A) -2 (B) 2 (C) 8 (D) -8

7. For what value of k is $f(x) = \begin{cases} x^2+k, & x \le 1 \\ 3x+2k, & x > 1 \end{cases}$ continuous at $x = 1$?

(A) 2 (B) 0 (C) -2 (D) -1

8. At $x = 5$, the function $f(x) = \dfrac{x^2-25}{x^2-4x-5}$ has

(A) a removable discontinuity
(B) a jump discontinuity
(C) an infinite discontinuity
(D) no discontinuity

9. The function f is continuous on $[0, 8]$. Selected values are shown.

x	0	2	5	8
$f(x)$	3	-4	1	-2

What is the least number of zeros of f in $[0, 8]$?

(A) 1 (B) 2 (C) 3 (D) 4

10. $\lim_{x\to 9} \dfrac{\sqrt{x}-3}{x-9} =$

(A) $\frac{1}{6}$ (B) $\frac{1}{3}$ (C) 0 (D) does not exist

11. If $\lim_{x\to 4} f(x) = 2$ and $\lim_{x\to 4} g(x) = -1$, then $\lim_{x\to 4} \dfrac{3f(x)-[g(x)]^2}{f(x)+g(x)} =$

(A) -5 (B) 7 (C) $\frac{5}{3}$ (D) 5

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12. A function f satisfies $4 - x^2 \leq f(x) \leq 4 + x^2$ for all x. Then $\lim_{x \to 0} f(x)$

(A) is 0
(B) is 4
(C) is 8
(D) cannot be determined from this information

13. $\lim_{x \to 3^-} \dfrac{x+1}{x-3} =$

(A) 4 (B) ∞ (C) $-\infty$ (D) 0

14. For the function q graphed in question 2, $\lim_{x \to 1^+} q(q(x)) =$

(A) 2 (B) 3 (C) 0 (D) 1

15. The graph of $f(x) = \dfrac{x-1}{x^2-1}$ has a vertical asymptote at

(A) $x = -1$ only
(B) $x = 1$ only
(C) $x = -1$ and $x = 1$
(D) no value of x

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Part B · Graphing calculator required

16. To three decimal places, $\lim_{x\to 0} \dfrac{e^{2x}-1}{\sin x}$ is

(A) 1.000 (B) 0.500 (C) 2.000 (D) 0.035

17. To three decimal places, $\lim_{x\to 0} \dfrac{1-\cos(3x)}{x^2}$ is

(A) 9.000 (B) 4.500 (C) 1.500 (D) 0.000

18. Selected values of a function g are shown.

x	1.9	1.99	2	2.01	2.1
$g(x)$	4.61	4.9601	1	5.0401	5.41

The best estimate of $\lim_{x\to 2} g(x)$ is

(A) 1 (B) 4.96 (C) nonexistent (D) 5

19. To three decimal places, $\lim_{x\to 1} \dfrac{\ln x}{x-1}$ is

(A) 1.000 (B) 0.000 (C) 2.718 (D) 0.368

20. A function h is continuous on $[1,4]$ with $h(1)=-2.3$ and $h(4)=1.7$. Which statement must be true?

(A) h is increasing on $[1,4]$.
(B) $h(2.5)=-0.3$.
(C) $h(c)=0$ for some c in $(1,4)$.
(D) h has exactly one zero in $(1,4)$.

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